MID-TERMEXAMINATION

(CBCS)

Course Name:<B.Tech: AI&ML/ CSE-AI >, Semester:<2nd>

(May, 2023)

Subject Code: BAI - 102	**Subject: Object Oriented Programming using JAVA**
Time : 1 ½ Hours	**Maximum Marks : 30**

Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1		**(2.5*4)**	
	(a) Explain JVM, JRE and JDK with the help of diagram.		
	(b) Explain the use if 'this' in java.		

	(c) Find the 5 errors in the following code
	public class MIDSEM{
	public void Main (String [] args){
	char s[] = "abc"
	system.out.println(s);
	}
	}		

	(d) What is an abstract class? Give brief example.		
Q2		**(5,5)**	
	(a) Write a java program that Inputs marks and prints corresponding grade 85-100: A, 70-85: B, 60-70: C, 50-60: D, and<50:FAIL		
	(b) Create a class Room (RoomNo, type, cost). Create 2 constructors 1) Room(), 2) Room(RoomNo, type, cost). Create 3 room objects and show the concept of constructor overloading.		
Q3		**(10)**	
	Differentiate between a class variable, instance variable and local variable. WAP to create a class Employee having EmpNo, Name, Dept, Address. Use class variables to automatically assign unique employeenumbers whenever a new employee object is instantiated. Create the records for 3 employees by reading the details from the user and print them. Find out the class/instance/local variables in your program.		
Q4		**(10)**	
	(a) Differentiate between Adhoc and Pure Polymorphism.		
	(b) Createa super class Vehicle (Company, color, price, readV(),print()), derive classes 1) Cars(Name, Model, readC(),print()) and 2) Bike(Name, readB(), print()). Override theprint functions of derived classes to print the details of 2-2objects of Car and Bike type each.		